

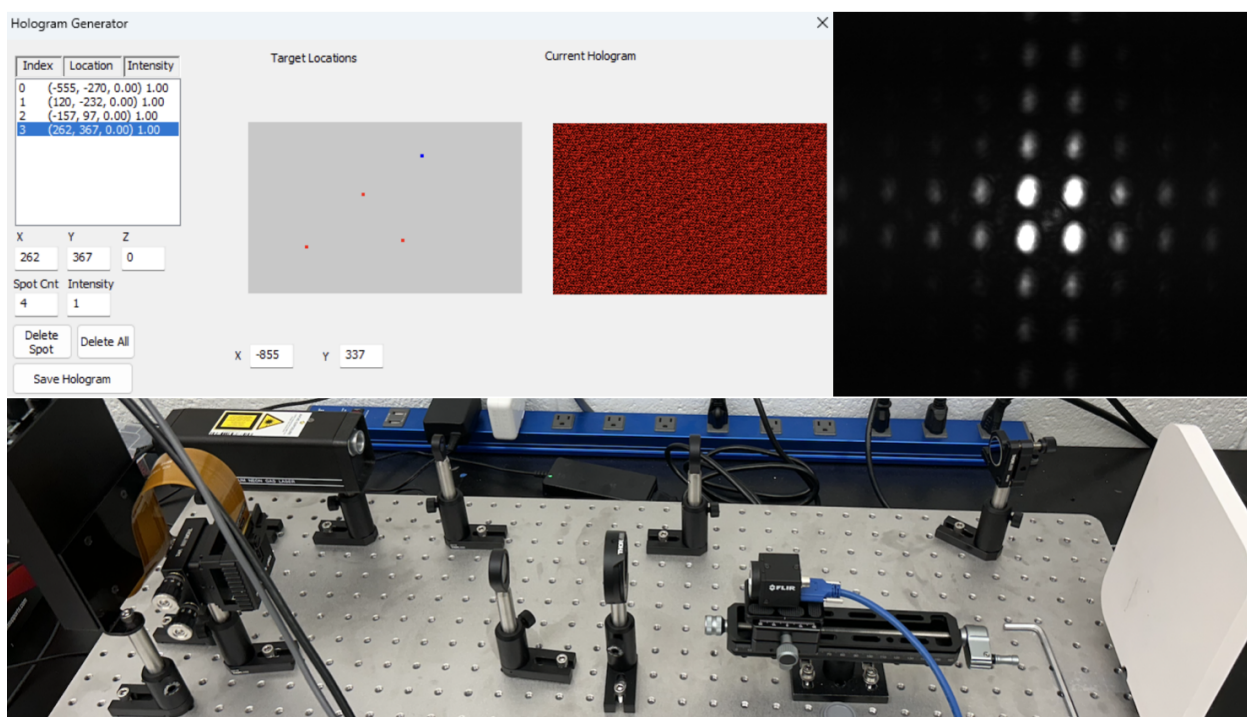


Physics

UNIVERSITY OF TORONTO

Advanced Undergraduate Laboratory

Digital Holography



Revisions

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1. Introduction

To bridge electronic data with optical systems, researchers in optical information processing have developed devices that convert electronic (or incoherent optical) signals into spatially modulated coherent light. These devices are known as spatial light modulators (SLMs) [1]. This is done by manipulating the orientation of liquid crystals inside a glass container so that incoming light is modified in a certain way, whether through phase or amplitude modification [2]. Each “pixel” (region of space on an SLM screen) of an SLM has a liquid crystal that has an effective refractive index. By sending a controlled voltage source to some of these pixels so that the SLM matches what is displayed on a computer, it becomes possible for light hitting each pixel of the SLM to be manipulated in different ways. One pixel could block all light like a black-colored pixel while another might allow light through like a white-colored pixel. Such devices can be used in complicated laboratory equipment like a multi-tron intrapulse interference phase scanner or in something used in everyday life like a video projector.

The goal of this experiment is to expose students to the optical physics behind holography via the usage of an SLM. The operation of an SLM is not as simple as shining a beam of light at an SLM. There are other optical factors that need to be considered, such as the polarization of the incoming coherent laser, the radius of the coherent laser as it hits the SLM, and making sure all of the lens focal lengths and distances are properly set. There is also the difficulty that while one can in principle generate any image via an SLM holography set-up, it is not as simple as, say, uploading a picture of a dog. Instead, one needs to input a transformation of that image of a dog so that the actual output from the SLM is the original dog image. By exposing students to all of these subtle difficulties in operating an SLM holography apparatus properly, they can better understand the intricacies to making SLM holography system work, and in so doing, perhaps inspire some students in the future to create SLM-inspired designs of their own.

The lab manual begins with an overview of the background theory and fundamental concepts of Spatial Light Modulators, followed by a step-by-step guide for setting up the software to generate basic computer-generated holography patterns. After establishing this foundation, students will explore quantitative measurements, such as diffraction gratings, to deepen their understanding beyond simple observation. Finally, students are encouraged to pursue more advanced analyses using complex patterns or their own innovative ideas among many.

2. Theory

The theory section begins with an overview of digital holography, focusing on the core apparatus used in this experiment, the spatial light modulator (SLM). It then introduces various types of phase patterns that can be generated using the SLM software, including periodic phase patterns, axial and radial phase patterns, and custom-designed patterns.

2.1 Spatial Light Modulators (SLM)

In 1948, Dennis Gabor proposed an imaging technique known as wavefront reconstruction, now widely recognized as holography. The core challenge in this process is to record and then reconstruct both the amplitude and phase of an optical wave originating from a coherently illuminated object [1]. A hologram is an image generated by manipulating a coherent light source, such as a laser, which maintains a constant frequency, wavelength, and phase. Spatial Light Modulators (SLMs) are programmable devices capable of manipulating the intensity, phase, and polarization of an incoming light wave on a pixel-by-pixel basis [2]. For instance, to create an image where the left side appears dimmer than the right, the SLM can be programmed to reduce the amplitude of the incident light on the left side while increasing it on the right, resulting in a corresponding brightness gradient.

An LCoS (liquid crystal on silicon) SLM referred to Figure 1 manipulates light through a protected liquid crystal layer [3]. The liquid crystal layer consists of a series of rod-shaped materials that are birefringent, meaning that its refractive index depends on the polarization of light. The silicon layer consists of silicon components each individually controlled by a voltage source. The idea is that by inputting voltage into some particular part of the silicon layer, a corresponding liquid crystal rod can be rotated. Due to the birefringent property of liquid crystals though, that means that the refractive index of this activated liquid crystal rod is different from the other liquid crystal rods. In doing so, light polarization can be manipulated to create a holographic image [4].

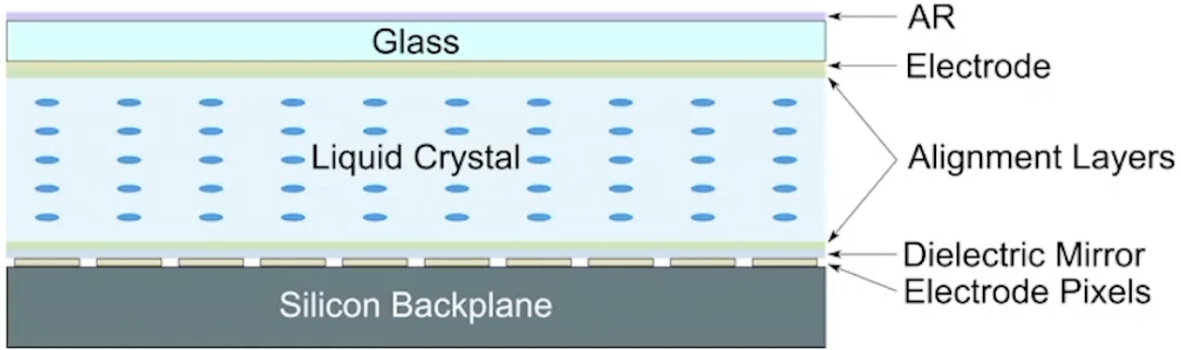


Figure 1: A diagram of the LCoS screen structure [5].

As an example, we shall consider phase modulation [6]. We can first consider a no-voltage situation where the liquid crystals are parallel to the glass layer of the SLM. This means incoming light will diffract most strongly, thus producing a larger output pattern. However, by applying a voltage source, we can rotate the liquid crystals so that they are oriented perpendicular to the SLM glass layer. Due to this, the diffraction of the light as it contacts the liquid crystals, and thus its phase, will change so that it will diffract less strongly. Thus, the output pattern will be much more concentrated.

2.2 Diffracted Patterns

When light incident on the diffraction grating, each groove acts as a secondary source for scattering the light. One can easily observe from Figure 2a, that the geometrical path difference between the light diffracted by successive grooves are $d \sin \theta_i - d \sin \theta_d$, where d is the grating spacing. Constructive interference arises when the optical path difference between light waves scattered from adjacent grooves equals an integer multiple of the wavelength. As a result, specific angles of diffraction emerge where the intensity of light is maximized, based on the principle of interference. The famous property of gratings is expressed in a simple equations, called the grating equations [7]:

$$\sin \theta_m = \sin \theta_i + m \frac{\lambda}{d}, \quad m = 0, \pm 1, \pm 2, \dots, \quad (1)$$

Refer to Figure 2b, where θ_i and θ_m denote the angles of the incident and m^{th} diffracted waves with respect to the normal of the grating surface. Here, λ is the wavelength of the light, and d is the grating period. In the case of the SLM, the grating period can be interpreted as the number of pixels over which the pattern repeats.

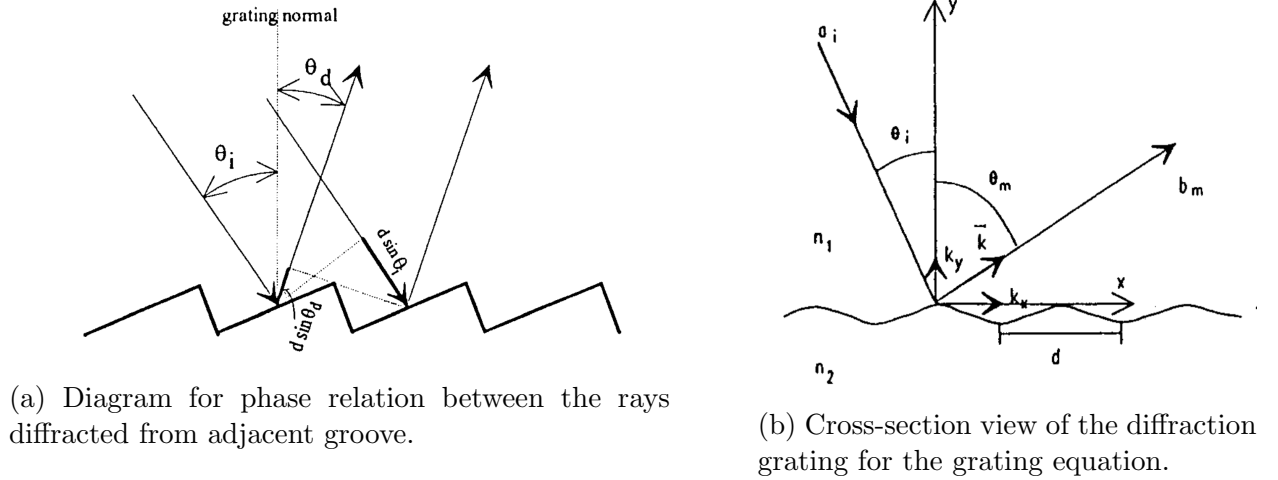


Figure 2: Diffraction Grating Diagram [7]

When a lens of focal length f is used to observe this pattern at its back focal plane, the spatial position x_m of the m^{th} diffraction order on the camera (or screen) is:

$$x_m = f \tan \theta_m \quad (2)$$

Under the small-angle approximation ($\theta_m \ll 1$), we can use:

$$\tan \theta_m \approx \sin \theta_m \approx \frac{m\lambda}{d} \quad (3)$$

Substituting into Equation 2, we get:

$$x_m \approx f \cdot \frac{m\lambda}{d} \quad (4)$$

This spacing is independent of the grating type (binary, sinusoidal, blazed), and depends only on the system parameters: wavelength λ , grating period d , and focal length f . LCoS SLMs are specifically designed to modulate the phase of incoming light by altering the refractive index, with negligible absorption. As a result, they are not intended for amplitude modulation. In other words, all the diffraction gratings observed in this experiment will function as reflective phase gratings.

2.2.1 Binary Diffraction

A binary phase grating is a periodic optical element that modulates the phase of light in a two-level stepwise fashion, alternating between two phase values, typically 0 and π . This is one of the simplest forms of a phase-only grating, often used in SLM-based experiments due to its straightforward implementation.

The phase modulation of a one-dimensional binary grating with period d can be modeled as a square wave:

$$\phi(x) = \begin{cases} 0, & 0 \leq x < \frac{d}{2} \\ \pi, & \frac{d}{2} \leq x < d \end{cases} \quad (\text{repeated periodically}) \quad (5)$$

Because the grating is periodic, we can express its transmittance function as a Fourier series [7]:

$$t(x) = \sum_{m=-\infty}^{\infty} c_m e^{i2\pi mx/d} \quad (6)$$

For a binary phase grating alternating between $+1$ and -1 , the Fourier coefficients c_m are nonzero only for *odd* diffraction orders (i.e., $m = \pm 1, \pm 3, \dots$) and are zero for even orders:

$$c_m = \begin{cases} 0, & m \text{ even} \\ \frac{2}{\pi m}, & m \text{ odd} \end{cases} \quad (7)$$

This means the diffraction pattern consists of symmetric bright spots at odd orders, and no light appears at even-order angles (including the 0th order). However you can try with different phase values for the strips, and an example will be shown in Figure 3.

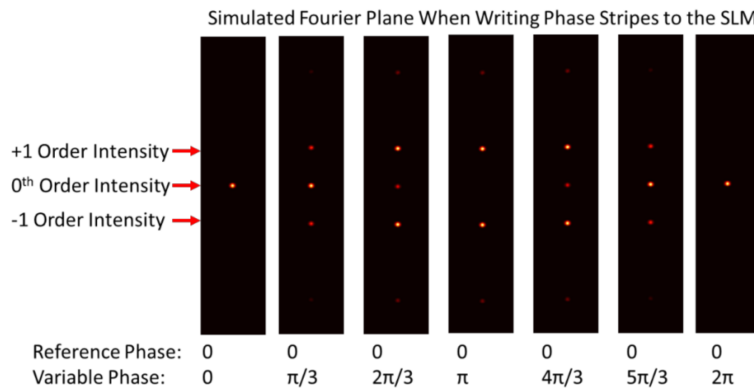


Figure 3: Changing phase value for binary gratings [6].

2.2.2 Blazed and Sinusoidal Diffraction

The blazed grating features a linear phase ramp that increases smoothly from 0 to 2π over each period, then resets. This shape resembles a sawtooth wave, as illustrated in Figure 4a. The continuous phase ramp causes constructive interference to concentrate light primarily in a single direction, resulting in most of the optical energy being directed into the $+1^{\text{st}}$ diffraction order, while other orders are significantly suppressed or absent.

In contrast, a sinusoidal grating modulates the phase in a smooth sine wave pattern across each period, as shown in Figure 4b. Unlike the abrupt phase jumps seen in binary or blazed gratings, the gradual phase transition of a sinusoidal grating distributes light more evenly across multiple diffraction orders. As a result, the diffraction pattern is symmetric, with observable bright spots appearing in both positive and negative orders.

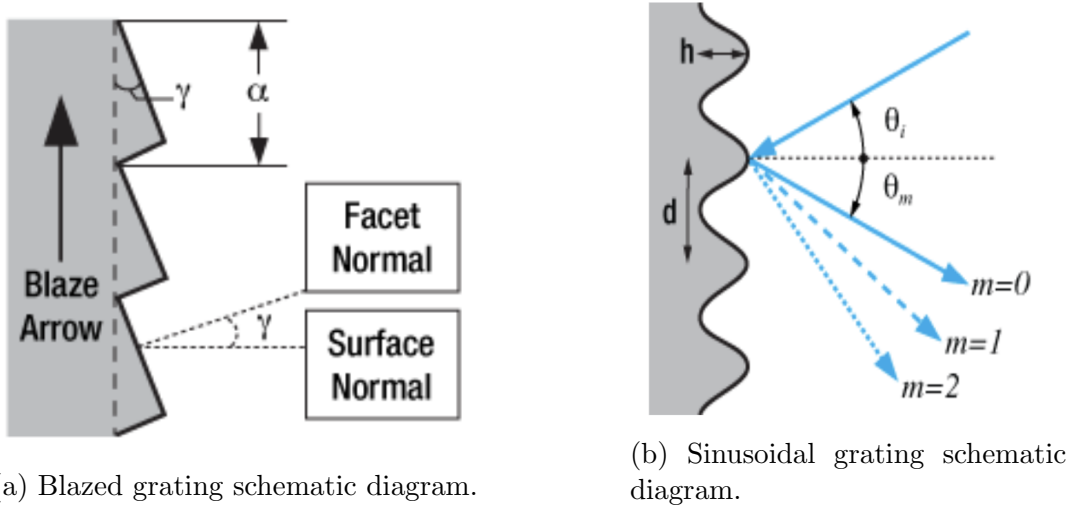


Figure 4: Blazed and sinusoidal grating pattern schematic [8]

2.2.3 Fresnel Lens

Fresnel lenses are thin, lightweight alternatives to traditional lenses, which use concentric grooves to bend light toward a common focus [9]. A cross section view of the physical lens can be found in Figure 5.

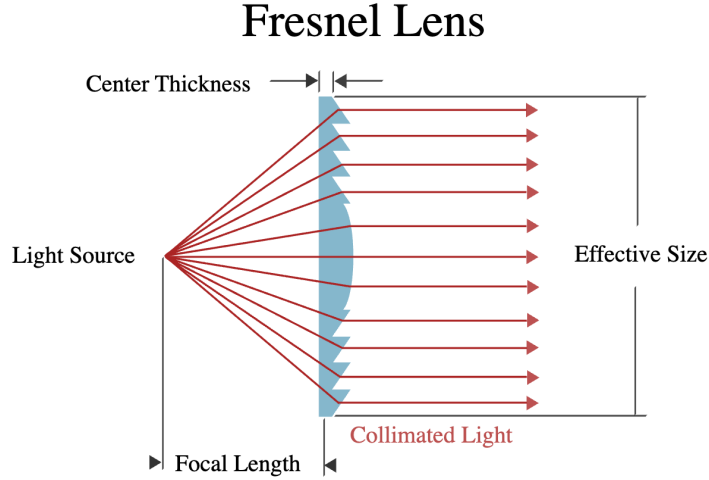


Figure 5: Profile view of the cross section of physical fresnel lens [9].

We can describe the ideal phase delay of a thin lens as:

$$\phi(x, y) = -\frac{\pi}{\lambda f}(x^2 + y^2) \quad (8)$$

The Fresnel lens exhibits a distinct radial pattern, determined by the focal length f and the wavelength of the incoming light λ . When implemented on a Spatial Light Modulator (SLM), the Fresnel lens is represented as a digitized version of the ideal phase function and operates purely through diffraction. The SLM displays a quadratic phase profile that modulates the wavefront of the reflected beam, effectively focusing it. This phase-only modulation does not rely on physical curvature but instead achieves focusing through controlled constructive interference [10].

2.3 Non-diffracted Beams

Unlike conventional Gaussian beams that spread as they propagate, non-diffracted beams maintain their intensity profile over long distances without significant diffraction. Well-known examples include Bessel beams, Airy beams, and other engineered variants. The SLM software provides a user friendly GUI that allows one to easily generate approximations of Bessel beams for experimental use.

2.3.1 Bessel Beam

The Bessel function is the precise solution to the Helmholtz equation in wave form under cylindrical coordinates [11]. The complex amplitude of an n^{th} -order Bessel beam can be expressed as:

$$E_n(r, \phi, z) = Ae^{ik_z z} J_n(k_r r) e^{\pm in\phi} \quad (9)$$

where $J_n(k_r r)$ is the n^{th} order Bessel function, and it describes the transverse profile (the plane perpendicular to the direction of propagation) of the beam, such as Figure 6.

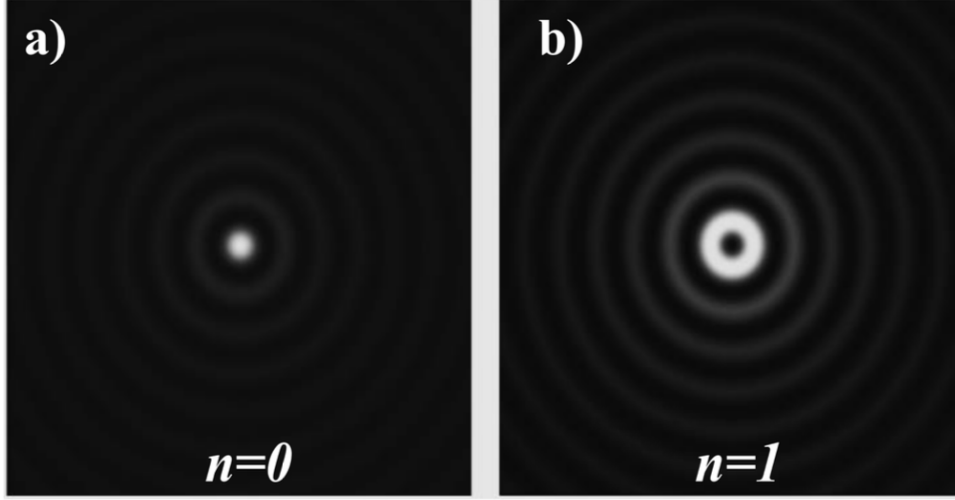


Figure 6: The left side is the transverse profile of an ideal Bessel beam intensity plot with zero-order Bessel function, and the right side is the first-order Bessel beam [11].

However, as one can observe, the Bessel beam tends to oscillate indefinitely as $r \rightarrow \infty$. This implies that an ideal Bessel beam would extend infinitely in space, requiring infinite power to realize [11]. In practice, however, a good approximation of a Bessel beam can be produced in the laboratory.

Creating Bessel beam recently with SLM has become an efficient way, as it's capable of producing Bessel beams in all orders with careful calibration. Phase pattern like axicons, concentric rings, vortex, and the BEST bessel beam phase pattern [12], they all serve as a good approximation or approach of the ideal Bessel beam.

3. Apparatus

3.1 Physical Apparatus

The physical apparatus for this experiment is mounted on a large optical breadboard in room MP 227. All components are securely fixed in place with their positions and angles pre-aligned, so further adjustments are likely unnecessary. A red HeNe laser (632.8 nm, 0.8 mW, linearly polarized) provides the coherent beam for the setup. The laser is linearly polarized in the vertical direction, matching the polarization requirements of the SLM. The beam first passes through a pair of lenses with focal lengths of 25 mm and 200 mm, expanding its diameter by a factor of eight. It is then reflected by a mirror onto a Meadowlark *XY Phase Series SLM* (1920×1200 resolution, pixel pitch of $9.2\mu\text{m}$), which features vertically aligned liquid crystals (slow axis). The SLM is connected to the lab computer via HDMI and USB cables.

Light reflected from the SLM spans a range of diffraction angles. The zeroth-order component is selected and focused by a 200 mm lens, then passes through a linear polarizer before reaching a CCD camera. The external polarizer, placed after the SLM, projects the modulated polarization onto a fixed axis, thereby controlling the measured intensity. The camera is mounted on a translational stage, allowing the observation plane to be adjusted along the optical axis. Since the lens placed after the SLM has a focal length of 200 mm, a meter ruler is used as a rough distance reference. However, precise alignment should be achieved by observing the focus of the diffraction pattern and adjusting the camera position accordingly. Both the SLM and the camera are controlled via software on a connected computer. A photo of the apparatus (excluding the computer) is shown in Figure 7.

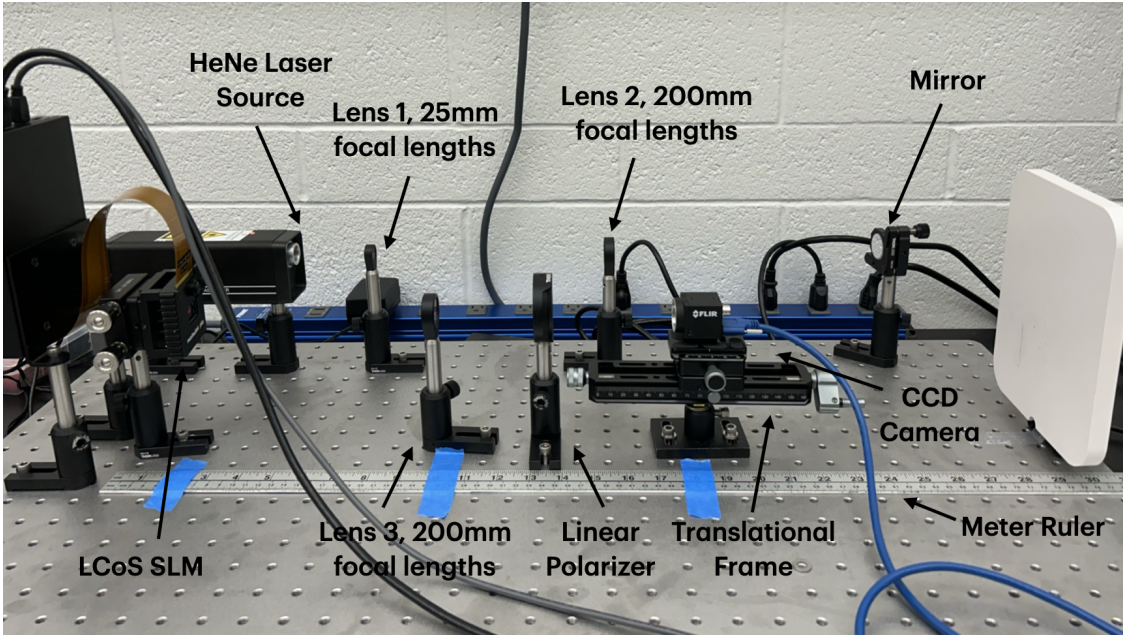


Figure 7: Optical setup on the lab breadboard with all components labeled, including the beam-expanding lens pair, SLM, polarizer, and CCD camera.

3.2 SLM Software

To display different patterns on the SLM and operate it as a "digital holography" setup, use the lab software located at:

`C:\Program Files\Meadowlark Optics\BlinkHdmi.exe`

Double-click to open the program (avoid opening it multiple times, as this may cause it to fail), and you will see an interface similar to Figure 8.

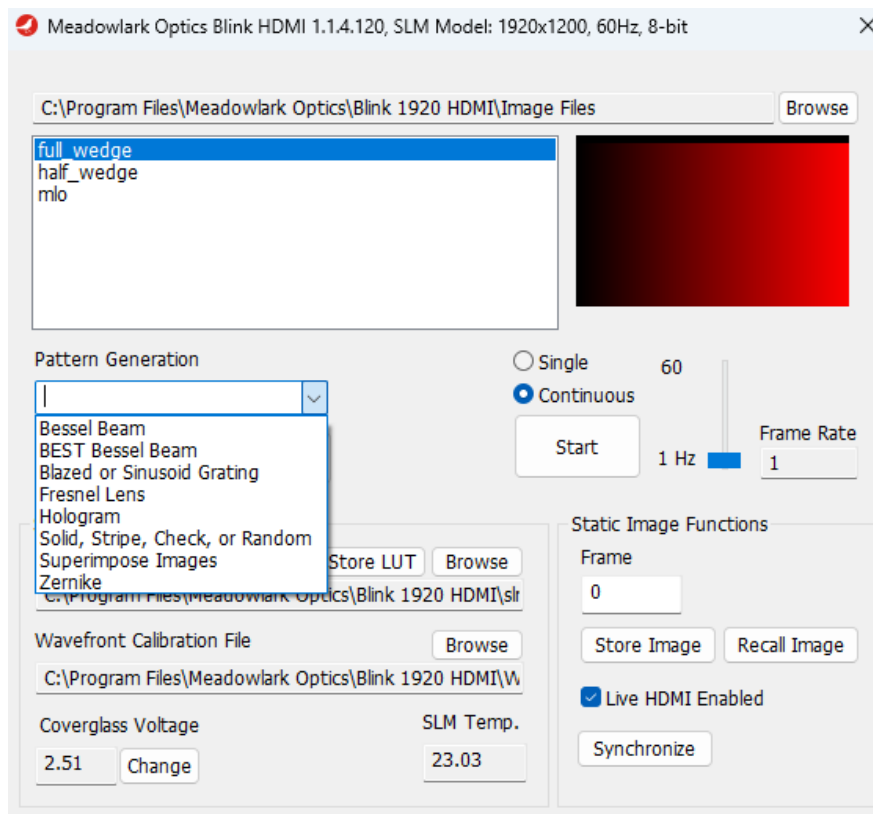


Figure 8: SLM software GUI, with the pattern generation drop-down menu opened.

The drop-down menu under "Pattern Generation" includes many predefined options. You can manually adjust the settings within each option or load saved bitmap files. Additional instructions for tuning pattern parameters and performing advanced operations can be found in later sections or in the user manual provided with the apparatus [6] (*1920 HDMI User Manual.pdf*).

3.3 Camera Software

To control the camera, open the software at desktop **SpinView**, and choose to load `USB Interface 0 -- Blackfly S BFS-U3-120S4M`

To avoid overexposure, set the exposure auto mode to *Off* and manually adjust the exposure time to $10\ \mu\text{s}$. Similarly, disable the gain auto mode and set the gain to 0 dB. You

may need to slightly adjust the mirror, optical path, or the camera's translational stage to bring a clear, bright spot to the center of the screen, as shown in Figure 9.

To save the current frame, click the blue button with the downward arrow. You can also record a short video clip using the adjacent red button. Note that both exposure time and gain may require fine-tuning depending on the lighting conditions and your specific setup.

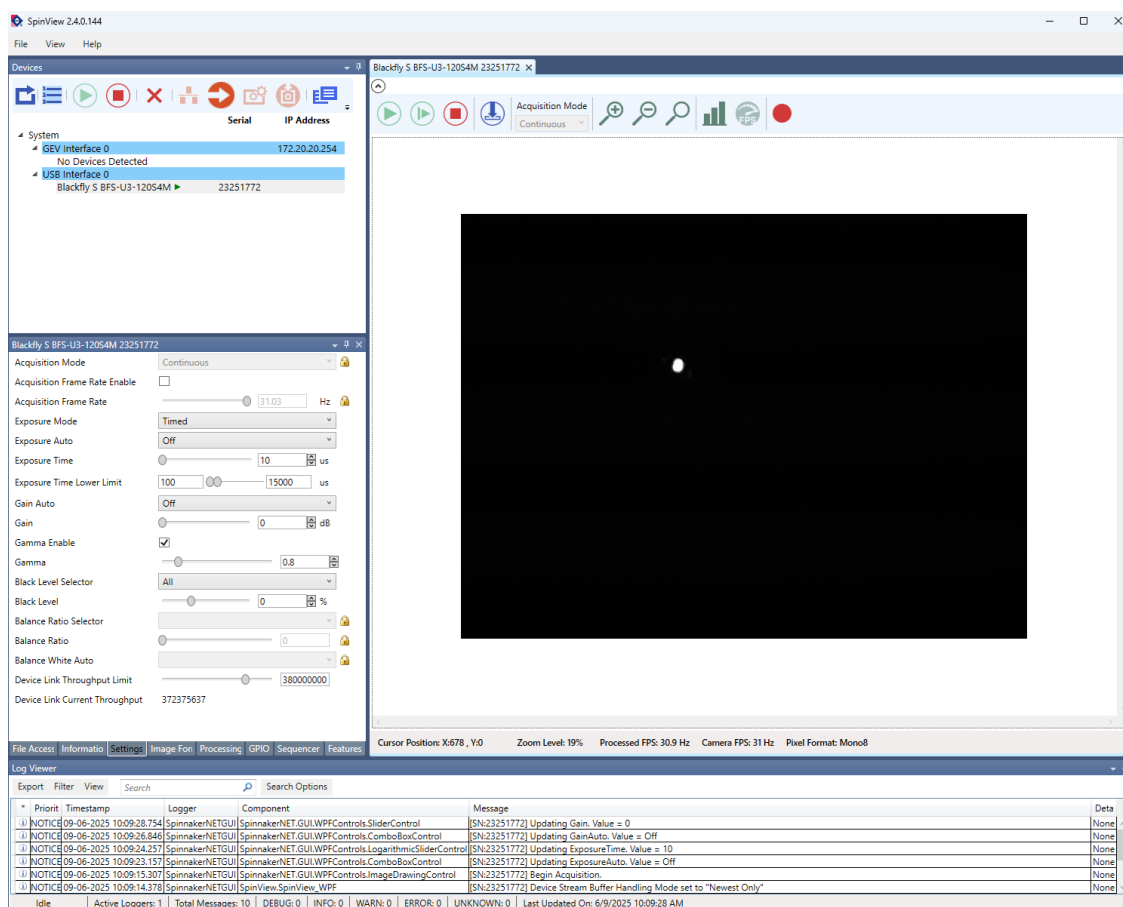


Figure 9: camera software GUI after adjusting the exposure, with a clear bright spot.

Safety Reminder

- **Laser Safety**

One of the components of the experimental apparatus is a class 3R laser. This class of lasers includes laser pointers and lasers used for fiber optic applications [13]. They have low power of less than 5 mW, but can still cause harm if one stares into the beam [13]. Therefore, one must take proper safety precautions by avoiding direct exposure to the laser beam [13]. Caution must also be taken with reflections of the beam mirrors and other such reflective surfaces. It is recommended to complete U of T's Closed Beam and Low Classes Laser Safety course [13] before starting work on the experiment.

- **Thermal Hazard**

Some of the components do get hot as they are used, specifically the camera and the SLM. Fortunately, if they are touched, the heat is more of a nuisance rather than something extremely dangerous (the camera though does get hotter than the SLM). However, this will be the one time the projector is mentioned, since there is an extreme temperature hazard with its lamp component that is so problematic that it needs to be addressed. The lamp the projector uses has a power consumption of 200 watts. In comparison, a house light bulb has a power consumption of about 60 watts, and they are already hot to the touch. As such, significant caution **MUST** be exercised when using the projector due to the lamp.

- **Optics Maintenance**

Some of the components are susceptible to smudging, scratching, or possibly breaking, specifically glass materials such as those for mirrors, lenses, cameras, and the SLM. Some of these come with a glass-cover, such as the SLM and the camera, while others could be manipulated with plastic gloves to prevent smudging. If not taken into account, smudging can be a mild nuisance, and breaking something can be more concerning, but of most importance is to be careful with the SLM since it cost about \$10,000 and took about four months to acquire.

NOTE: *This is not a complete list of all hazards; we cannot warn against every possible dangerous stupidity, e.g. opening plugged-in electrical equipment, juggling cryostats, etc. Experimenters must constantly use common sense to assess and avoid risks, e.g. if you spill liquid on the floor it will become slippery, sharp edges may cut you, etc. If you are unsure whether something is safe, ask the supervising professor, the lab technologist, or the lab coordinator. If an accident or incident happens, you must let us know. More safety information is available at <http://www.ehs.utoronto.ca/resources.htm>.*

4. Experimental Procedure

The experimental procedure outlines the overall flow of the experiment. You are expected to complete Sections 4.1 and 4.2, followed by a selection of topics from Section 4.3 for in-depth analysis. Each section may include a brief tutorial or set of instructions for guidance, along with a list of questions to help you get started.

4.1 Basics of the Apparatus

4.1.1 Question List:

- Why does the SLM require the incoming light to be polarized? Consider the role of the liquid crystal layer in modulating the light.
- Explain the purpose of each component in the setup. Specifically, why are a lens and a polarizer placed after the SLM?

4.2 Camera Calibration and Diffraction Grating

4.2.1 Gratings

Figure 10a shows the interface for generating blazed and sinusoidal grating patterns, while Figure 10b displays the interface for strips and other binary grating patterns. The parameter *Period* in Figure 10a and *Pixels per strip/check* in Figure 10b represent the same quantity: the spatial period of the grating in pixels. This period determines the physical grating spacing d according to Equation 10:

$$d = n \times \text{pixel pitch} \quad (10)$$

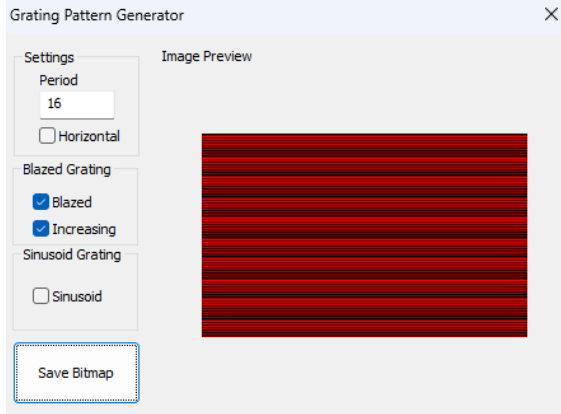
where n is the user-defined period in pixels, and the pixel pitch of our SLM is $9.2 \mu\text{m}$.

The parameters *Pixel Value One* and *Pixel Value Two* define the two phase levels used in binary grating patterns. These values are entered in RGB grayscale units, where a value of 0 or 256 corresponds to a full 2π phase shift. Therefore, the relationship between grayscale value and phase is linear, with one unit corresponding to approximately $\frac{2\pi}{256}$ radians.

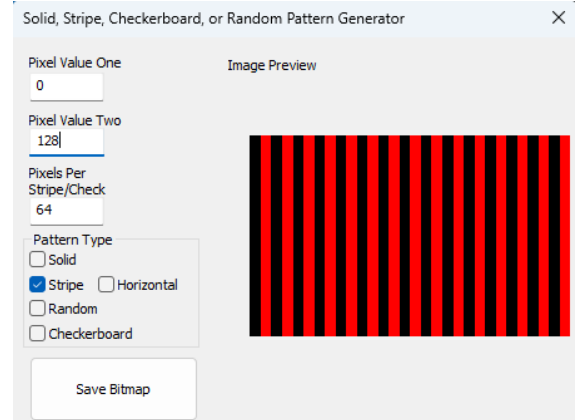
For example, to create a grating with a phase shift of $\frac{\pi}{3}$, the corresponding grayscale value would be:

$$\frac{256}{2\pi} \cdot \frac{\pi}{3} = \frac{256}{6} \approx 43$$

Thus, you can set the pixel value to approximately 43 to achieve the desired phase difference.



(a) Blazed and sinusoidal pattern generation GUI.



(b) Strips and binary gratings pattern generation GUI.

Figure 10: Grating pattern generation GUI

4.2.2 Fresnel Lens

This interface in Figure 11 allows users to generate a digital Fresnel lens pattern by specifying parameters such as the number of waves of defocus, lens radius, and center position in pixels. Increasing the Defocus value adds more waves of phase curvature, effectively reducing the focal length. The Radius sets the extent of the lens, while Center X and Center Y define the optical center. Checking Cylindrical or Horizontal produces line-focus patterns instead of circular lenses.

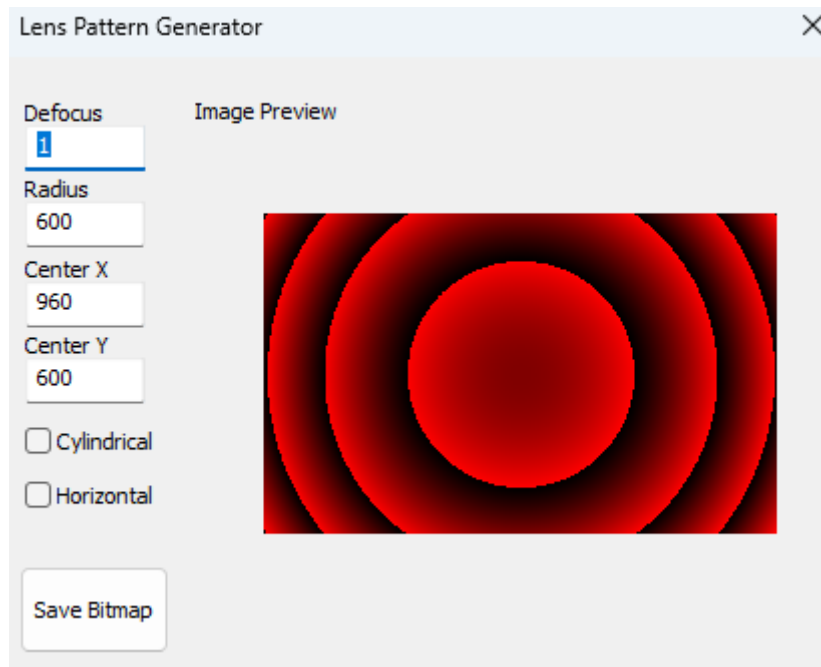


Figure 11: Fresnel lens GUI.

4.2.3 Question List

- Calculate the pixel-to-physical-distance ratio of the camera, where you are expected to generate a known diffraction pattern and compare with number of pixels in the image. You may use any image processing software for this purpose. Be sure to include an uncertainty measurement in your calculation.
- What differences do you observe among the three grating patterns? Discuss the causes of these differences in terms of the phase pattern geometry and diffraction efficiency.
- Experiment with different settings besides strips using the *Solid, Strip, Checkerboard, or Random Pattern Generator*. Describe and explain how each pattern results in the observed diffraction pattern on the camera.
- What are the near-field and far-field approximations in diffraction? How do they differ in terms of calculation and interpretation? Adjust the translational position of the camera to observe how the diffraction patterns change with distance.
- What is the significance of a Fresnel lens in diffraction patterns? Does it operate under a near-field or far-field approximation?
- Measure the focal length of the Fresnel lens pattern generated by the SLM. Does it introduce more uncertainty compared to a physical lens?

4.3 In-depth pattern analysis and further ideas

A list of in-depth pattern analyses and optional extensions you can explore in this lab is provided below. You are encouraged to develop your own ideas, do literature reviews or use any of the suggested topics as a starting point for further investigation.

- **Bessel Beam:** The Bessel beam is a fundamental example of a non-diffracting beam and can be generated using both the *Bessel Beam Phase Pattern* and *BEST Bessel Beam Phase Pattern* interfaces. The latter incorporates advanced engineering techniques as described in [12]. To begin, you are encouraged to experiment with the Vortex, Axicon, and Concentric Rings patterns available in the standard interface. Once familiar with these, the *BEST Bessel Beam* GUI can be used to generate optimized Bessel-like beams for deeper analysis. A general overview of Bessel beams can be found in reference [11].
- **Hologram Encode:** The Hologram GUI allows you to create phase profiles that generate multiple focal points in the Fourier plane using a Weighted Gerchberg-Saxton algorithm [14]. This algorithm runs on the GPU and performs an iterative FFT-based optimization to compute the hologram. You can click on the preview window to add focal spots, then adjust their x, y, z coordinates and intensities manually. Using this GUI, you can create custom 2D or 3D arrays of focal spots—such as grids, letters, or volumetric patterns [6]. You can control the relative brightness of each spot, explore how focal depth influences resolution, and test how different iteration settings affect hologram quality.

- **Superimpose Images:** The Superimpose Image function in the Meadowlark SLM software allows you to combine two phase patterns—for example, adding a Fresnel lens to a phase hologram or incorporating wavefront correction into an existing pattern. While it does not directly generate images in the Fourier plane like holography, it enables more complex beam shaping by merging multiple phase functions into a single display pattern on the SLM.
- **Zernike:** Zernike polynomials are a set of continuous, orthogonal functions defined over a unit circle. Since many optical systems involve circular apertures or pupils, these polynomials are widely used to represent and analyze optical wavefronts. The SLM software includes an intuitive interface for generating various Zernike phase patterns. Refer to [15] as a starting point to explore and apply different Zernike modes in your experiments.
- **Expand Laser Beam Size:** Currently, the laser beam occupies only a small portion of the SLM’s active area, which limits the efficiency and resolution of the modulation. To improve performance, consider expanding the beam using appropriate optical components such as a beam expander or a pair of lenses. Consult the lab technicians for assistance in selecting and aligning these elements properly.
- **Generate Non-diffracted Phase Pattern:** As mentioned in Section , there are several types of non-diffracting beams. The apparatus manual [6] also introduces how to write custom code to generate your own patterns on the SLM. It’s worth exploring this by programming additional non-diffracted phase patterns and testing their efficiency with the setup.
- **Adaptive Optics:** Adaptive optics refers to the correction of image distortions by compensating for aberrations in the incoming wavefront. For instance, astronomers use adaptive optics to correct distortions in starlight caused by atmospheric turbulence, improving image clarity [16]. One method of implementing this correction involves using a Spatial Light Modulator (SLM). A representative study demonstrating this application is provided in references [17] and [18].
- **3D Pattern Reconstruction Using a Translational Frame:** By translating the camera along the optical axis, you can capture a series of 2D diffraction patterns at different depths. These images can be used to reconstruct the 3D propagation of the optical field, revealing how structured beams—such as Fresnel or Bessel beams—evolve in space. Consider how to build a 3D reconstruction system using a 2D camera.
- **Generate Arbitrary Image:** The final idea demonstrates the principle of digital holography, where you can generate an arbitrary image, such as a dog. However, directly displaying the image on the SLM will not reproduce it at the output due to diffraction effects, similar to how a diffraction grating produces an interference pattern rather than a physical replica. To recover the desired image, you must input its Fourier transform onto the SLM, so that it reconstructs correctly in the far field.

References

- [1] J. W. Goodman. *Introduction to Fourier Optics*. Roberts and Company Publishers, Englewood, CO, 3rd edition (2005). ISBN 9780974707723.
- [2] J. Seagrave. “Understanding the Jargon of LCOS Spatial Light Modulators (SLMs)” (2024). URL <https://photonics.laser2000.co.uk/blogs/understanding-the-jargon-of-lcos-spatial-light-modulators-slms/>.
- [3] J. Wang *et al.* “Optical Fiber Communications, Chapter 12 - Multimode Communications Using Orbital Angular Momentum, Section 12.3.1.3: Spatial light modulator (SLM)”. In *Optical Fiber Communications* (2023). URL <https://www.sciencedirect.com/topics/computer-science/spatial-light-modulator>.
- [4] Y. Yang, Forbes A., and L. Cao. “A review of liquid crystal spatial light modulators: devices and applications”. *Opto-Electron. Sci.*, 2(8):230026 (2023). doi:10.29026/oes.2023.230026.
- [5] Thorlabs. “EXULUS Spatial Light Modulators – Principles and Applications”. YouTube video (2021). Available online: <https://www.youtube.com/watch?v=iX994oL1H-M> (accessed June 2025).
- [6] Meadowlark Optics. *User Manual 1920 x 1152 XY Phase Series and 1920 x 1200 XY Phase Series Spatial Light Modulator With HDMI Controller* (2024).
- [7] E. G. Loewen and E. Popov. *Diffraction Gratings and Applications*. Marcel Dekker, New York (1997). ISBN 9780824793098.
- [8] “Diffraction gratings tutorial”. Accessed June 17, 2025, URL https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=9026.
- [9] MeetOptics. “Fresnel Lenses”. <https://www.meetoptics.com/academy/fresnel-lenses#what-are-fresnel-lenses> (n.d.). Accessed: 2025-06-23.
- [10] O. Mendoza-Yero *et al.* “Diffraction-Based Phase Calibration of Spatial Light Modulators With Binary Phase Fresnel Lenses”. *J. Display Technol.*, 12(10):1027–1032 (2016). URL <https://opg.optica.org/jdt/abstract.cfm?URI=jdt-12-10-1027>.
- [11] S. N. Khonina, N. L. Kazanskiy, S. V. Karpeev, and M. A. Butt. “Bessel Beam: Significance and Applications—A Progressive Review”. *Micromachines*, 11(11) (2020). ISSN 2072-666X. doi:10.3390/mi11110997. URL <https://www.mdpi.com/2072-666X/11/11/997>.
- [12] R. Lu *et al.* “Video-rate volumetric functional imaging of the brain at synaptic resolution”. *Nat. Neurosci.*, 20(4):620–628 (2017). doi:10.1038/nn.4516.
- [13] University of Toronto. “Closed Beams and Low Classes Laser Safety” (2024).

- [14] Y. Wu *et al.* “Adaptive weighted Gerchberg-Saxton algorithm for generation of phase-only hologram with artifacts suppression”. *Opt. Express*, 29(2):1412–1427 (2021). doi:10.1364/OE.413723. URL <https://opg.optica.org/oe/abstract.cfm?URI=oe-29-2-1412>.
- [15] V. Lakshminarayanan and A. Fleck. “Zernike polynomials: a guide”. *J. Mod. Optic.*, 58(7):545–561 (2011). doi:10.1080/09500340.2011.554896. <https://doi.org/10.1080/09500340.2011.554896>, URL <https://doi.org/10.1080/09500340.2011.554896>.
- [16] “ALPAO Webinar - Understanding the basics of Adaptive Optics”. <https://www.youtube.com/watch?v=3I8bdMCYmpQ>. Accessed Jan. 26, 2025.
- [17] P. S. Salter and M. J. Booth. “Adaptive optics in laser processing”. *Light Sci.Appl.*, 8(1):110 (2019). doi:10.1038/s41377-019-0212-6. Published: 29 November 2019, URL <https://www.nature.com/articles/s41377-019-0215-1>.
- [18] K. Yamamoto *et al.* “Interactive eye aberration correction for holographic near-eye display”. In *Proceedings of the Augmented Humans International Conference 2021*. ACM (2021). Accessed Jan. 26, 2025, URL <https://dl.acm.org/doi/10.1145/3458709.3458955>.